Technology and Philosophy of AI

Instructor

Dr Clinton Staley "Clint"

CS Professor Emeritus, Cal Poly SLO

UATX STEM Professor

805 835-5024 for texts

Overview

We'll look at several aspects of AI in the class. For each I've given some relevant readings, all from current websites. There's no paper textbook for the class, since the area moves too fast. You'll also want a phone or laptop connected to Prin's wireless so you can try out a few AI websites during the class.

All readings are entirely optional. There are no tests or homework in this class :). And even the phone or laptop is up to you. I'll display website results on screen during lecture, and you're welcome to share a phone or laptop with the person next to you.

Getting Set Up

One of the most-requested items in the survey I sent out was some hands-on use of AI. We'll do this on the first day of class, trying out three different "LLM"s (aka chatbots, like ChatGPT). This will be easier if you've already visited the three sites below, each of which offers a different LLM, and tried out one prompt at each. Again, this is *not required*, but if you have time, it's good prep.

So, the set up steps are:

1. Visit the website for "Claude" -- a chatbot offered by Anthropic. https://claude.ai/chats It'll ask you for an email and send a confirmation code for you to enter, but there's no other requirement for registration. After that, you should reach a web page where you can have a conversation with Claude.

Example prompt: Please describe, in 200 words, the new UATX university in Austin Texas

or if you want something more creative, ask it to write a short story based on a title. You might use Hemingway's famous 6-word short story "For sale. Baby shoes. Never worn", but ask for something more positive than what is implied by the title.

Example prompt: Write a 200 word short story, based on the title "For sale. Baby shoes. Never worn." but avoid themes of loss or childlessness.

3. Do the same for ChatGPT, perhaps the most famous of these LLMs. It's at chatgpt.com. Registration is similar -- just supply an Email. No need to pay for the premium version; the free version will suffice for our class. Try out our two prompts and see if it does a better job than Claude.

4. Two LLMs to compare and contrast is enough, but if you're interested, you can try Gemini from Google at gemini.google.com which has a similar free registration setup. See if it does a better job with the prompts. And the site Duck AI offers a choice of LLMs, smaller and less resource-intensive than the main ones, and with anonymous access (no account or login needed).

5. Try asking ChatGPT to create an image for you, based on a description you supply. This will show how AI works in media other than text.

6. And finally, for fun you might check out the best recent work in AI generation of movies via an AI system called Sora

Readings

Here are the suggested readings and/or exercises for each of the five areas of AI we hope to explore.

Hands-on AI and AI Basics

We'll look at how to effectively prompt an LLM. Here is a quick online reference with suggestions. Spend maybe 15 minutes getting a feel for the strategies suggested. See if you can improve on the prompts from above and get better results.

> SuperAnnotate Prompting Hints

If (and only if) you'd like more depth, this lengthier page may be worth a look.

> TDS Prompting Hints

AI History

Our class will include an overview of AI history. Wikipedia has a good summary article. Ignore the "Precursors". Just start at "Birth of Artificial Intelligence". There's a lot of detail; free free to skim. But, keep an eye out for the pattern of AI hype followed by periods of "AI winter", as far back as the 1960s. Note how far back the idea of AI was discussed. It's not a new concept. And, watch for the two major approaches to AI that competed across the decades.

> Wikipedia: AI History Overview

AI and Society

One of our sessions will discuss the near term (next decade or so) impacts of AI in various areas.

The first is AI and copyright. Since LLMs are trained on books, news articles, etc that are copywritten, does this mean they're stealing others' work, or are they just doing what you and I do -- learning from others and then putting ideas into our own words. Take a look at these brief articles, and then we can talk about them in class.

> AI and Copyright

Federalist Society Review of USPTO Postion

Another big question has to do with liability. If a chatbot gives bad medical advice, or a self-driving car hits someone, who is legally liable? The user of the AI? The creator of it? AI is known to be imperfect, so is it fair to sue if you've been warned about this? These Wall Street Journal and Wired articles summarizes the situation well.

WSJ on AI Liability
Wired on AI Liability

Perhaps the biggest question about near term effects of AI is how it will change employment. Will AI provide new jobs to replace lost ones, like prior tech revolutions, or is this time different? What careers will be most affected? How should a new entrant to the labor market (like you, after graduation) best prepare?

These articles offer a variety of takes, ranging from pessimistic to optimistic.

MIT Review: AI and Employment (traditional optimistic)
CBS: Jobs Likely to be Replaced (semi-pessimistic)
Warpnews: AI Creates New Middle Class? (thoughtful)
AI and Coding Jobs (Pessimistic)
Anthropic CEO's Warning (Pessimistic)

AI Technology

Many of you were interested in a layman's understanding of how AI works. We'll devote a class session or so to this. Even in an upperdivision computer science class, the basics can take a couple of weeks, so we're not going to cover them at a deep mathematical level. (Those really interested are welcome to get together with me for lunch some time this week for a serious "geek" discussion :))

But, at least an impression of the math can be gained from several good Youtube videos. Here's one:

Technical Overview of Deep Learning

And there are also these excellent explanations by 3Brown1Blue, perhaps the finest explainer of mathematical concepts on the web today. Don't worry about following all the equations unless you really want to, but try to get a feel for what happens in "deep learning" via the diagrams. And you don't need to watch all of them. Take in the first one, and if you feel like more, try the others.

Deep Learning 1
Deep Learning 2
Deep Learning 3
Transformers 1

We'll cover several AI areas that are *not* LLMs or transformers. The first is a Convolutional Neural Network or CNN. This is a deep learning network like an LLM, but designed to imitate the function of the eye. This article gives a brief summary of the idea.

Overview of CNNs

Reinforcement learning (RL) is another area of AI that deserves attention. The first article covers the basics. No need to read the "Markov Decision Process" math too carefully. You can skim that latter half. The second article shows a critical application of RL called AlphaFold. Watch the brief video at the start of it, and read the accompanying text if you'd like to -- the latter is optional.

> Intro to Reinforcement Learning
> AlphaFold

If you are interested in a deep dive on the leading technical edge of AI, from two world class experts speaking with a good interviewer, I'd recommend this podcast

> Fridman Podcast on AI Trends

AI Philosophy

In one of our sessions, we'll address the long term possibility of human-level AI, and whether such AI should be regarded as conscious or having rights. A central thought-experiment in this is Searle's Chinese Room argument, summarized well in this reference. Pay attention not only to the argument, but to the counterarguments offered.

> Chinese Room Argument

And here is a talk by Searle at Google. At least listen to the first couple minutes of the talk, just to get a feel for what Searle is like. The rest is optional.

> *Searle Speaking at Google*

And finally, here is a good MIT article discussing AI rights.

> AI Rights (light discussion)